

Yun-Tang Chang

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Work Experience

Software Engineer (Backend Intern)

Sep 2023 - Jun 2025

NCKU Office of International Affairs (OIA)

- Built the backend for an agentic assistant serving 500+ students, with a custom MCP server for document retrieval.
- Optimized frequent-query handling with FAISS-based semantic caching, improving end-to-end throughput by 3×.
- Designed Redis-coordinated asynchronous worker pipelines to scale concurrent agent workflows.

Research Experience

Research Intern

Jun 2026 - Present

Department of ECE, Computer Engineering Group, University of Toronto

- Developing autoscaling strategies for I/O-sensitive stateful operators in the Flink Kubernetes Operator autoscaler.
- Modeling CPU-memory tradeoffs of stateful operators to guide autoscaling policies under constrained cluster resources.

Undergraduate Researcher

Feb 2026 - Jun 2026

Secure System Lab, National Taiwan University

- Integrated vsock with eBPF sockmap to delegate guest networking in SEV-SNP CVMs, reducing networking TCB by 10×.
- Optimized the vsock networking path, improving throughput by **21.8%** and HTTPS serving by **9.2%** over virtio-net.
- Profiled kernel-level vsock behavior with eBPF and perf to trace and eliminate performance bottlenecks.

Projects & Open Source

duckdb-query-condition-cache

- Implemented a predicate cache in DuckDB's filter-pruning path to accelerate repeated dashboard queries.
- Sped up queries by up to **3.2×** on the HDFS_v2 log dataset by skipping decompression of non-qualifying data blocks.

duckdb-table-inspector

- Built DuckDB storage observability tooling to analyze database file usage and table-level compression efficiency.
- Reached 16K+ DuckDB community downloads and helped investigate free-block inflation and index-reclaiming issues.

moonlink

- Contributed to **moonlink**, a Rust engine streaming data into Apache Iceberg for real-time lakehouse updates.
- Implemented a GCRA-based rate limiter for throughput emulation and improved ingestion reliability through enhanced error handling across REST APIs and metadata operations.

Education

National Cheng Kung University (NCKU)

Taiwan

Bachelor of Science in Computer Science and Information Engineering

Sep. 2022 - Jun. 2026

- **Academics:** 4.19/4.3 GPA

Awards & Hackathons

2025 AWS Generative AI Hackathon – Taipei, Taiwan

Apr 2025

- Designed a retail analytics pipeline using Jetson devices, AWS Kinesis, and SageMaker for real-time heatmap inference.
- Integrated Lambda, S3, and API Gateway to support edge-to-cloud data streaming and analytics delivery.

Taipei Hackathon 2024, 3rd Place – Taipei, Taiwan

Sep 2024

- Engineered **ParkFlow**, a real-time parking and navigation app integrated into Taipei's **400,000-user** city platform.
- Reduced lookup latency by **53%** through geospatial search and database-query optimization.
- Implemented the system with Python, Vue 3, Flutter, and geospatial indexing over the Taipei Parking API.

Skills

Languages: C, C++, Python, Go, Rust

Systems / Data / Infra: Linux, eBPF, Redis, PostgreSQL, DuckDB, Apache Flink, Docker, Kubernetes, GitHub Actions

AI / Cloud: FAISS, AWS Kinesis, SageMaker, Lambda, S3